MODULE 10 · DOMAIN 1 - PEOPLE

Problem solving and team decisions

Finding root causes instead of treating symptoms, and choosing the right way for a group to decide.

DOMAIN I - People EXAM WEIGHT 33% MODULE 10 of 49

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Most problems in a scenario are symptoms. The right answer treats **the cause**.

A simple problem-solving sequence

01

Define

State the problem precisely, with facts.

02

Find the cause

Analyse why it happens, not just what.

03

Generate options

Several ways to address the cause.

04

Decide

Choose, using an agreed method.

05

Act and check

Implement, then verify it worked.

ON THE EXAM

Skipping straight from 'define' to 'act' is the most common error, in real projects and in exam options.

Symptom or root cause?

TREATING THE SYMPTOM

Fast, and it comes back

- Fix each defect as it appears
- Add overtime when late
- Separate people who argue
- Add another approval step

TREATING THE CAUSE

Slower, and it stays fixed

- Find why defects are created
- Find why estimates are wrong
- Find what the argument is really about
- Find why errors get through

THE TIEBREAKER

A recurring problem means the cause is still there. Look for the option that finds it.

Five whys: an example from Project Lantern

- Why did the booking app crash at go-live? **The appointment service timed out.**
- Why? **It was flooded with requests from the kiosks.**
- Why? **The kiosks retry every second when they get no reply.**
- Why? **The retry setting was never tested under load.**
- Why? **Load testing did not include the kiosks.** That is the cause to fix.

> ON THE EXAM
>
> Keep asking why until you reach something you can change in the process. The number five is a guide, not a rule.

Two more root-cause tools

- **Cause-and-effect diagram**, also called fishbone or Ishikawa: groups possible causes by category.
- Categories are often people, process, technology, materials, environment and measurement.
- **Pareto analysis**: rank causes by frequency. Often a few causes explain most of the problems.
- Use them **with the team**. The people doing the work usually know the causes.

> ON THE EXAM
>
> Root-cause analysis is a team activity. The project manager facilitates; the team supplies the knowledge.

Ways a group can decide

METHOD	WHAT IT MEANS	FITS WHEN
Unanimity	Everyone actively agrees	Rare; very high-stakes, small groups
Consensus	Everyone can support it, even if it was not their first choice	Commitment matters; the usual team default
Majority	More than half agree	Speed matters; the minority can accept it
Plurality	The largest group wins, even under half	Many options, low stakes
Autocratic	One person decides, perhaps after input	Emergencies, or it is clearly their decision

ON THE EXAM

Consensus is not unanimity. It means everyone can live with and support the decision.

Quick techniques teams use

TECHNIQUE	HOW IT WORKS	GOOD FOR
Fist of five	0 to 5 fingers of support	Checking consensus, surfacing concerns
Roman voting	Thumbs up, sideways or down	A quick yes, unsure or no
Dot voting	Dots placed on options	Prioritising a longer list
Delphi	Anonymous expert rounds, summarised	Estimates where experts sway each other
Weighted criteria	Score options on weighted criteria	Complex choices, such as suppliers

ON THE EXAM

Pick the technique by purpose: support, priority, avoiding influence, or criteria.

Fist of five in practice

- **5**: fully support, will champion it. **4**: support. **3**: can live with it.
- **2**: have concerns to discuss. **1**: serious concerns. **0**: cannot support it.
- Anyone showing **2 or fewer** explains their concern, and the team discusses it.
- Vote again. Proceed when **everyone shows 3 or more**.

> ON THE EXAM
>
> The low votes are the point. They surface concerns that a show of hands would bury.

Diverge, then converge

- **Diverge**: generate many options without judging them.
- **Converge**: evaluate, combine and choose.
- Start with **silent, individual** idea generation, then share. It stops the first idea anchoring everyone.
- Criticising ideas while generating them shuts the quiet voices down.

> ON THE EXAM
>
> Separate generating from judging. Mixing them produces fewer and safer ideas.

Groupthink and the loudest voice

- **Groupthink**: a group agrees quickly to keep harmony, and nobody tests the idea.
- The **highest-paid person's opinion** can end a discussion before evidence is heard.
- Counter them: invite dissent, assign a devil's advocate, run a pre-mortem, gather input anonymously.
- A **pre-mortem** asks: 'It is a year from now and this failed. Why?'

> ON THE EXAM
>
> Fast, unchallenged agreement on something important is a warning sign, not a success.

When to decide

- **Reversible** decisions: decide quickly with the information available, and adjust later.
- **Irreversible** decisions: take more care, gather more evidence, involve more people.
- Lean thinking's **last responsible moment**: delay a commitment until just before waiting would cost you.
- Weeks of debate over an easily reversed choice is waste. Timebox it.

> ON THE EXAM
>
> Match the effort to the reversibility. Cheap-to-change decisions deserve speed, not perfection.

Record and communicate decisions

- Keep a **decision log**: what was decided, why, by whom, and on what date.
- Share decisions with everyone affected, including those who were not in the room.
- Reopen a decision only when **new information** appears, and say what it is.
- Decisions that live only in memory get relitigated in every meeting.

> ON THE EXAM
>
> A decision nobody recorded will be made again. Log it, share it, and reopen it only for new facts.

Project Lantern: a team decision

- Question: should clinics **go live one at a time**, or three at once?
- The team **diverged** silently, listing risks and benefits, then shared.
- A **pre-mortem** surfaced a risk nobody had said aloud: support staff stretched across three sites.
- **Fist of five**: two people showed a 2. After discussion, one clinic at a time won support from all.

> ON THE EXAM
>
> The method surfaced the decisive risk. A quick majority vote would have chosen three at once.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A problem keeps recurring	Root-cause analysis with the team
Need to check team support quickly	Fist of five or Roman voting
Experts sway each other's estimates	Delphi: anonymous rounds
Instant agreement, nobody challenges	Invite dissent; pre-mortem
Teams interpret the vision differently	Break down where understanding diverged
Weeks of debate on a reversible choice	Timebox it and decide

ON THE EXAM

Fix causes, not symptoms, and choose a decision method that fits the purpose.

> For four sprints in a row, the team has found similar integration defects late in the sprint. Each time, developers fix them individually, working late.
>
> The team is tired, and the product owner is losing confidence in the release date.

What should the project leader do?

A Facilitate a root-cause analysis of the defects with the team.

B Add a senior developer's review before any code is merged.

C Request an extra tester for the rest of the release.

D Ask the developers to fix defects faster so the sprint is not affected.

◆ PAUSE AND DECIDE

What should the project leader do?

A Facilitate root-cause analysis.

Rung 2 with the team: similar defects every sprint means a cause nobody has found.

B Add a senior review.

A new gate that catches symptoms later, and creates a bottleneck.

C Extra tester.

Finds the defects, but does not stop them being created.

D Fix faster.

The symptom, again, with more pressure on a tired team.

THE SIGNAL

'Similar defects, four sprints running': a recurring problem. Find the cause before adding people or gates.

> A team has discussed changing its code branching strategy for twenty minutes. Most people seem positive, but two have said little.
>
> The project leader wants to check support quickly and make sure any concerns come out.

Which technique should the project leader use?

A. A majority vote by show of hands.

B. Decide personally, since most people seem positive.

C. Fist of five, discussing any vote of two or fewer.

D. The Delphi technique with external experts.

◆ PAUSE AND DECIDE

Which technique should the project leader use?

A. Majority vote.
Quick, but buries the two quiet people's concerns under the majority.

B. Decide personally.
A team working-practice decision, taken away from the team.

C. Fist of five.
Fast, checks support, and gives low votes a reason to speak.

D. Delphi with outside experts.
Slow, and aimed at expert estimates, not team consensus.

THE SIGNAL

Check support fast AND surface concerns: fist of five does both.

> A project manager needs duration estimates for a novel piece of engineering work. Five internal experts will estimate it.
>
> In past estimating sessions, the most senior expert spoke first and the others tended to adjust towards her figure.

Which approach should the project manager use?

A. Ask the most senior expert to provide the estimate, since the others defer to her anyway.

B. Hold an open meeting and average the figures the experts give.

C. Take a majority vote among the five experts.

D. Use the Delphi technique, with anonymous rounds summarised by a facilitator.

◆ PAUSE AND DECIDE

Which approach should the project manager use?

A The senior expert decides.

Throws away four experts' knowledge.

B Open meeting, average.

The anchoring still happens before anyone speaks their number.

C Majority vote.

Estimates are not a vote, and the influence problem remains.

D Delphi.

Anonymity removes the senior expert's pull; rounds still converge.

THE SIGNAL

Experts influencing each other: anonymity. That is exactly what Delphi is for.

> In a short meeting, a team agreed unanimously to select a supplier recommended by the sponsor's preferred contact. Nobody asked any questions.
>
> The contract is large, and switching suppliers later would be expensive.

What should the project manager do?

A Accept the decision, since it was unanimous.

B Before confirming, invite challenge, for example with a pre-mortem on the choice.

C Overrule the team and choose the supplier personally.

D Escalate the selection to the procurement board to decide.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept, it was unanimous.

Unanimous without questions, on an irreversible choice, suggests groupthink.

B Invite challenge; pre-mortem.

Tests an expensive, hard-to-reverse decision before committing.

C Overrule the team.

Replaces one untested view with another.

D Escalate to procurement.

Avoids the problem rather than testing the decision properly.

THE SIGNAL

Instant, unchallenged agreement on an irreversible choice: test it before you commit.

> Two agile teams are building parts of the same customer portal. They have delivered features that conflict: one assumes customers log in first, the other assumes they browse anonymously.
>
> Both teams say they followed the product vision.

What should the project manager do first?

A Choose one team's interpretation and ask the other to rework.

B Escalate the conflict to the sponsor to decide.

C Work with both teams to find where their understanding of the vision diverged, then restate it with examples.

D Give the teams separate product backlogs to avoid future conflicts.

◆ PAUSE AND DECIDE

What should the project manager do first?

A. Pick one interpretation.
Treats the symptom; the vision is still ambiguous for the next feature.

B. Escalate to the sponsor.
Too early. Nobody has yet found where the misunderstanding lies.

C. Find where understanding diverged; restate with examples.
The outline's own enabler: break down the situation to find the root cause.

D. Separate backlogs.
Hides the conflict inside one product, making it worse.

THE SIGNAL

Two teams, one vision, conflicting work: the vision is ambiguous. Find where, and fix the vision.

> A team has spent three weeks debating which user-interface component library to use. Both candidates are well supported.
>
> The developers agree that switching libraries later would take about two days.

What should the project leader do?

A Let the debate continue until the team reaches full agreement.

B Escalate the choice to the architecture board.

C Commission a detailed comparison study of the two libraries.

D Help the team timebox the decision, choose with what it knows, and revisit only if problems appear.

◆ PAUSE AND DECIDE

What should the project leader do?

A Debate until full agreement.

Three weeks already spent on a two-day decision. More is waste.

B Escalate to the architecture board.

Adds delay to a reversible, team-level choice.

C Detailed comparison study.

More analysis for a choice that costs two days to undo.

D Timebox, decide, revisit if needed.

Matches effort to reversibility, and keeps the decision with the team.

THE SIGNAL

Cheap to reverse, and already weeks of debate: timebox it and decide.

Find the cause. Choose a method that fits. Match the effort to the reversibility.

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